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Continuous polyethylene glycol precipitation of recombinant antibodies: Sequential precipitation and resolubilization

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ABSTRACT

A continuous precipitation-based capture step for recombinant monoclonal antibodies (mAbs) has been developed using polyethylene glycol (PEG) in a sequential precipitation approach. In the first step, impurities are removed by precipitation at low pH and low PEG concentrations. After removing the precipitate, mAbs are precipitated by pH adjustment and PEG supplementation. Yields of 86–94% and a reduction in host cell proteins (HCPs) to 7200–15,000 ppm were reached depending on the mAb. Host cell DNA was removed to 18–29 ng/ml, which was independent of the starting DNA concentration. The secondary structure and isoform distribution of mAbs are unaltered compared with the chromatographically purified reference material. The method was then shifted to a continuous operating mode in a single-use tubular reactor by converting the time to reach equilibrium into residence time. Tangential flow filtration (TFF) has been used as an alternative precipitate capture step to centrifugation. The yield was not compromised with the use of TFF (>92%), and impurity removal was even enhanced compared with centrifugation. Concentration of the mAb precipitate by TFF and washing by diafiltration preserves the floc structure of the precipitate, thereby accelerating the subsequent resolubilization.

1. Introduction

In recent years, polyethylene glycol (PEG) precipitation has received renewed interest, especially in the field of monoclonal antibody (mAb) purification [1–3]. PEG precipitation is a simple method that operates at ambient temperature and exhibits fast precipitation kinetics. Furthermore, PEG precipitation supports continuous production, an operating mode that is progressively being introduced into the biopharmaceutical industry, as it is more economical and permits the use of single-use equipment at a smaller scale [4].

In recent years, various kinds of precipitation and aqueous two-phase extraction systems have been investigated as alternatives to existing technologies such as chromatography [5–12]. PEG precipitation is the most prominent precipitation method for mAbs. A high-yield step that offers a relatively low selectivity, PEG precipitation is most often used as an initial capture step, which involves precipitation of the product, that is, mAbs or other molecules such as botulinum toxin B [1,2,9,13–15]. Giese et al. have also assessed PEG precipitation for impurity precipitation [16]. Product precipitation has the benefit of pool volume reduction during the precipitate resolubilization. However, it is technically more challenging than impurity precipitation, as the formed precipitate cannot simply be removed and discarded, but it has to be resolubilized. In classical PEG precipitation approaches that attempt to directly precipitate the product out of the clarified cell culture solution, especially large contaminants such as DNA are inadvertently coprecipitated with the product. Therefore, the burden on the subsequent purification steps is increased [13]. In order to reduce contaminants that would coprecipitate in the second step, an additional precipitation step can be introduced prior to the actual product precipitation. Caprylic acid precipitation and $CaCl_2$ flocculation have been investigated for this purpose [9,14,17,18]. Contaminant reduction can also be integrated into the cell harvest step by adding flocculants [19–21]. Although these approaches are feasible, their disadvantages complicate the process scheme and introduce additional substances into the process, which must be removed.

Most publications on PEG precipitation are focused on the actual product precipitation and parameter optimization. However, the crucial precipitate capture step is not addressed. Centrifugation is tacitly assumed to be suitable for this purpose. Solid–liquid

Abbreviations: CCCS, clarified cell culture supernatant; CD, circular dichroism; CHO, Chinese hamster ovary; CIP, cleaning in place; HCP, host cell proteins; HF, hollow fiber; HMWI, high molecular weight impurities; LMH, L/m^2/h; PBS, phosphate buffered saline; PEG, polyethylene glycol; TFF, tangential flow filtration; TMP, transmembrane pressure.

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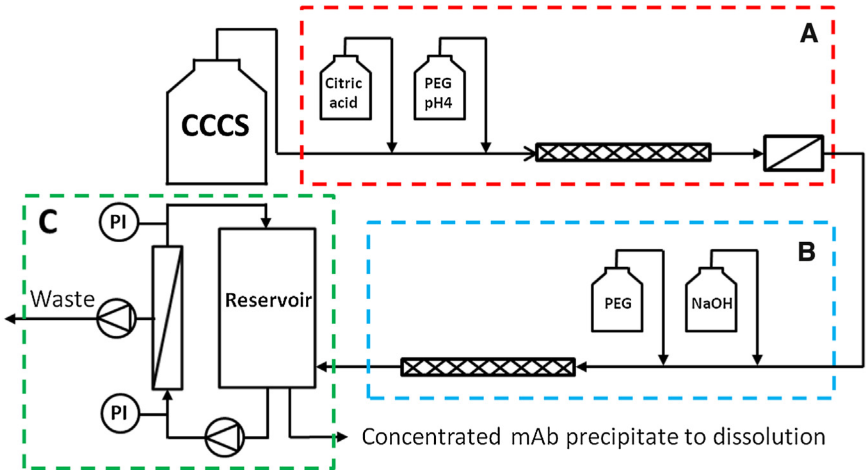

Fig. 1. Flowchart of the experimental setup for continuous sequential precipitation including discontinuous precipitate recovery by TFF. The clarified cell culture supernatant (CCCS) is supplemented by different stock solutions. Mixing is accomplished using helical-type static mixers. In the first step (A), impurities are precipitated and removed by depth filtration. In the second step (B), the mAb is precipitated. In the third step (C), the mAb precipitate is concentrated in the retentate before being dissolved.

Table 1
Titer, isoelectric points, and impurity levels of investigated IgG supernatants produced in CHO cell culture.

	mAb1	mAb2	mAb3
p*I*	9.3	8.2	7.6
Titer (mg/ml)	2.3	4.9	2.9
DNA (ng/ml)	5411	2278	806
HCP (ppm)	101,688	96,087	113,538

separation can be achieved using centrifugation because of the density difference of the precipitate and the mother liquor. However, although the method of choice for laboratory-scale experiments, centrifugation has some decisive drawbacks in the context of protein precipitation, especially at a large scale. During the centrifugation process, the precipitate flocs are compacted to a dense, paste-like mass. Resolubilization of the pellet is a slow process (in the order of hours) with a variability in the processing time. High capital investment, maintenance efforts, and low flexibility are additional disadvantages of centrifugation. Kuczewski et al. demonstrated the suitability of tangential flow filtration (TFF) as a precipitate capture step in a feasibility study of an extremely high-titer mAb [2]. The precipitate is first concentrated in the retentate, washed by diafiltration, and finally dissolved. Unlike in centrifugation, the individual flocs are preserved during TFF. These flocs exhibit rapid dissolution kinetics.

This work addresses some of the limitations of classical PEG precipitation. A sequential PEG precipitation-based capture step for mAb purification that solely relies on PEG addition and pH adjustment is presented. In a first step, impurities are precipitated and in the second step, the product is precipitated. This sequential approach circumvents the problem of coprecipitating DNA and shows increased host cell protein (HCP) removal. The starting point for the development of the sequential precipitation approach was the display of the lowest solubility of proteins at or near their *pI*, where the net charge is zero and repulsion therefore minimized. While the majority of CHO HCPs and DNA have isoelectric points in the acidic pH range, recombinant antibodies typically have isoelectric points at neutral or basic pH [22,23]. The sequential precipitation method was optimized for three recombinant antibodies of varying titers and isoelectric points (*pI*s) (see Table 1) in the batch mode; further, this was shifted to the continuous mode for one mAb (mAb3) as proof of concept. In addition, TFF with hollow-fiber (HF) modules was used as a precipitate capture step. TFF is a well-established technology with the advantages of a linear scale-up behavior and a single-use setup, thereby minimizing cleaning and validation efforts and eliminating the risk of cross contamination [24,25].

For feasible screening efforts in this study, the parameters for screening were kept to a minimum. Conductivity, PEG size, PEG concentration, and pH influence the PEG precipitation of proteins. The influence of conductivity was not screened for, as the aim of this study was to precipitate mAbs out of crude clarified cell culture supernatant (CCCS) with a predefined conductivity. It is well known that PEGs with higher molecular weight are more efficient up to about PEG6000, that is, lower concentrations are required for precipitation [26,27]. However, the higher efficiency is accompanied by higher viscosity [28]. Therefore, PEG6000 was selected, as it has been used in previous studies for precipitation at similar mAb titers with high efficiency and acceptable viscosities [1,9,14].

2. Materials and methods

2.1. Materials

2.1.1. Feed stocks

Clarified CHO cell culture supernatants of the three IgG_1 were provided by Novartis AG (mAb1 and mAb2) (Basel, Switzerland) and by Sandoz (mAb3) (Kundl, Austria). The supernatant characteristics are presented in Table 1. Supernatants were stored at −20 °C for the long term or at 4 °C for the short term.

2.1.2. Reagents

All chemicals were of analytical grade and were purchased either from Sigma–Aldrich (St. Louis, MO, USA) or from Merck Millipore (Darmstadt, Germany), unless otherwise stated.

2.2. Methods

2.2.1. PEG precipitation

PEG6000 stock solutions (45 w/v%) were prepared in 25 mM buffer and 150 mM NaCl, and these were set to the desired pH range of 3.5–8.5. The solubility curves of mAbs, the optimization of host cell DNA removal, and the mAb precipitation screening experiments were performed in 96-well DeepWell plates (Nunc, Roskilde, Denmark). The final working volume was 1.2 ml. The buffer, PEG6000 stock solution, and CCCS were mixed to screen different PEG concentrations at constant antibody concentrations. The

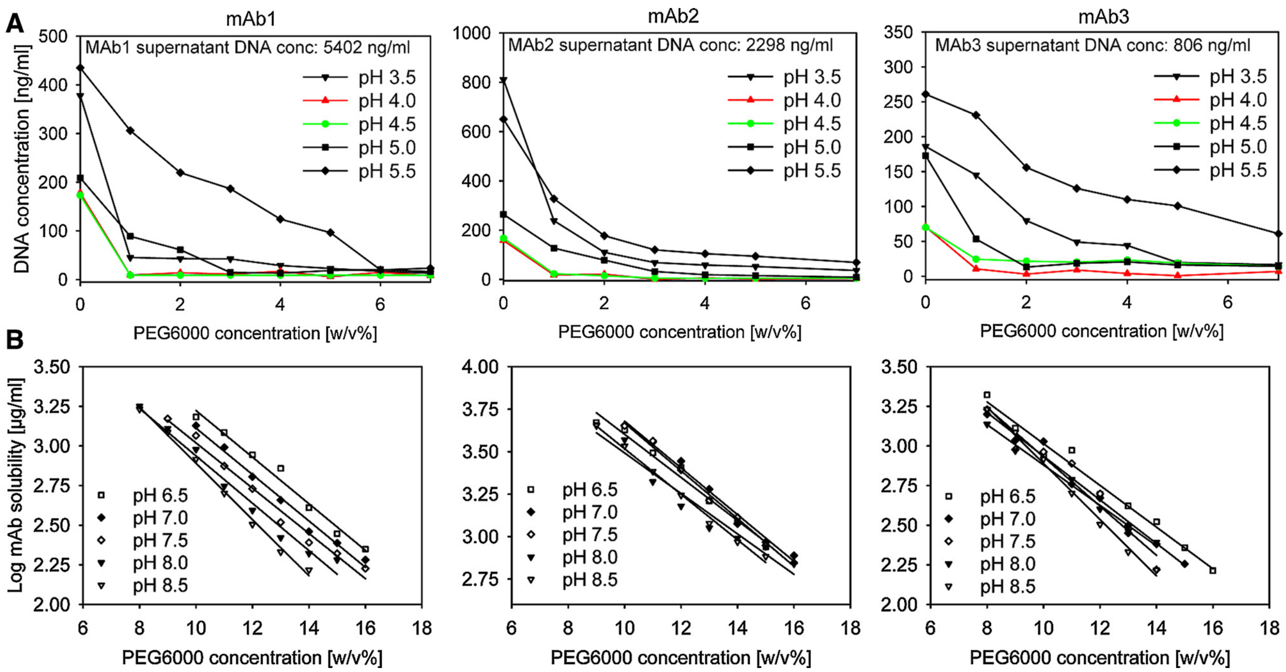

Fig. 2. (A) PEG precipitation screening in a pH range of 3.5–5.5 for DNA removal. The best performing conditions are marked in red and green (B) Solubility curves of the three mAbs at a pH range of 6.5–8.5. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

plates were sealed and incubated on a rotator (Stuart, Stone, UK) at 15 rpm for 2 h to reach equilibrium conditions. The precipitate was collected by centrifugation at 4000 × *g* for 20 min. The supernatants were discarded and pellets dissolved in phosphate-buffered saline (PBS). All experiments were performed at room temperature. Citric acid solution (3 M) was used to lower the pH to the desired level for removing contaminants, unless otherwise stated.

2.2.2. Continuous PEG precipitation

Continuous PEG precipitation was performed in a simple tubular reactor (ID: 4.8 mm, Tygon tubing R-3603 from IDEX, Wertheim, Germany) equipped with polypropylene Kenics-type, helical static mixers (Stamixco, Winterthur, Switzerland). The reactor length was 13.5 m to ensure a mean residence time of 15 min at a linear flow rate of 1.5 cm/s. PEG6000 stock solutions (45 w/v%) at the optimized precipitation pH were added to the CCCS, and 25% HCl and 10 M NaOH were used to adjust the pH. Peristaltic pumps (Reglo Digital, Ismatec, IDEX, Wertheim-Mondfeld, Germany) were used for pumping all solutions. Precipitated impurities were removed by 1.0/0.3 µm depth filtration in the first step (Sartoclear P, Sartorius, Göttingen, Germany). A schematic flowchart of the continuous precipitation setup is depicted in Fig. 1A and B.

2.2.3. Precipitate recovery by centrifugation

Antibody precipitate was recovered by centrifugation using a Multifuge X3-FR (Thermo Scientific, Waltham, MA, USA) at 3000 rcf for 15 min at room temperature. After centrifugation, the supernatant was discarded and the precipitate pellet was dissolved in phosphate buffered saline (PBS) if not stated otherwise. Pellets were dissolved on a rotator at 30 rpm overnight.

2.2.4. Precipitate recovery by TFF

A simple TFF system with minimal dead volume was custom-built. A Masterflex peristaltic pump (Cole-Parmer, Vernon Hills, IL, USA) was used as a recirculation/feed pump. A peristaltic pump (IPC, Ismatec, IDEX, Wertheim-Mondfeld, Germany) was used to control the permeate flux. The permeate flux was set by the permeate pump flow rate and measured by collecting the permeate on a scale (Sartorius, Göttingen, Germany). Feed, retentate, and permeate pressures were monitored using pressure sensors (PendoTECH, Princeton, NJ, USA). The TFF setup is depicted in Fig. 1C. The pressure and permeate flux data were collected using LabView (National Instruments, Austin, TX, USA). HF microfiltration cartridges (0.1-, 0.2-, and 0.45-µm pore size, and 110 cm^2 surface area) were obtained from GE Healthcare (GE, Uppsala, Sweden). HF membrane regeneration was performed on an Äkta Crossflow. The membrane was flushed with water for 60 min, followed by 60 min of cleaning in place (CIP) with 0.25 M NaOH. A water flux test was conducted to ensure membrane integrity. Sodium phosphate buffer set to the pH used for precipitation and supplemented with 14 w/v% PEG6000 (optimized PEG concentration for mAb precipitation) was used for system equilibration and diafiltration. Concentrated antibody precipitate was dissolved by dilution in PBS to about 10 mg/ml if not stated otherwise.

2.2.5. Protein A affinity chromatography

Amsphere protein A JWT203 resin (JSR Micro, Sunnyvale, CA, USA) was used for the preparative purification of all three antibodies on an Äkta Pure workstation (GE Healthcare, Uppsala, Sweden). PBS was used as an equilibration buffer. The column wash consisted of three steps (5 column volumes (CV) of equilibration buffer, 5CV of equilibration buffer supplemented with 1 M NaCl, and 5CV equilibration buffer). Elution was performed with 100 mM sodium acetate at a pH of 3.1. NaOH (250 mM) was used for CIP between runs. Four different buffers (sodium citrate, sodium acetate (both pH 5.3), sodium phosphate (pH 7.5), and Tris (pH 8.1)) at three conductivities each (2.5, 7.5, and 15.0 mS/cm) were tested for precipitate dissolution. Buffers of 50-mM concentration were prepared and set to the respective conductivities at 20 °C using either saturated NaCl (in the respective buffers) or HQ-H_2O.

2.3. Analytical methods

2.3.1. Protein A affinity chromatography

The mAb concentrations were determined using analytical protein A affinity chromatography, as described in Ref. [29], using an *n*-protein A CIMdisk monolithic column (BIA Separations,

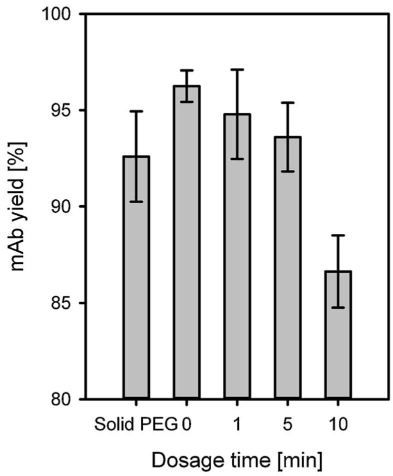

Fig. 3. Effect of dosage time of PEG addition on mAb yield for mAb3.

Ajdovcina, Slovenia). The equilibration and wash buffer consisted of 30 mM sodium phosphate and 1 M NaCl at pH 7.4 and the elution buffer was 10 mM HCl at pH 2.0.

2.3.2. Size-exclusion chromatography

High molecular weight impurities (HMWI) were determined by size-exclusion chromatography using a TSKgel G3000SWxl column (Tosoh Bioscience, Minato, Japan). Potassium phosphate (150 mM) at pH 6.5 was used as the mobile phase at a flow rate of 0.4 ml/min.

2.3.3. DNA concentration determination

The concentrations of double-stranded DNA (dsDNA) were measured with a Quant-iT Picogreen dsDNA assay kit (Life Technologies, Carlsbad, CA, USA), according to the manufacturer's instructions, in a 96-well format. In brief, 1:2 serial dilutions of samples and λ DNA standard were performed in a 96-well plate in $1 \times$ TE buffer (10 mM Tris–HCl, 1 mM ethylenediaminetetraacetic acid (EDTA), pH 7.5). Approximately 100 μl of each dilution was transferred to a fluorescent black-bottom 96-well plate, 100 μl of pre-diluted color reagent was added, and the signal intensities were measured in a plate reader. The excitation and emission wavelengths were 480 and 520 nm, respectively.

2.3.4. CHO HCP ELISA

Goat anti-CHO HCP antibody (3G-0016-AF, Cygnus, Southport, NC, USA) was used to coat MaxiSorp Immuno 96-well plates (NUNC, Roskilde, Denmark). Blocking was carried out by incubation with 3% bovine serum albumin (BSA, Sigma Aldrich) in Tris-buffered saline (TBS). A standard curve was prepared in duplicate with a CHO HCP standard (F553H, Cygnus). Samples and the pre-diluted CHO HCP standard were serially diluted at 1:2 with 1% BSA in TBS and transferred to the measurement plate. The goat anti-CHO HCP antibody conjugated to HRP (1:2,000 diluted in 1% BSA in TBS + 0.05% Tween 20) was used as the detection antibody. The TMB Peroxidase EIA Substrate Kit (Bio-Rad, Hercules, USA) was used for staining. The enzymatic reaction was stopped with 1 N sulfuric acid. The absorbance at 450 nm was measured using a Tecan F500 Infinite plate reader (Tecan, Maennedorf, Switzerland).

2.3.5. Circular dichroism spectroscopy

Samples were buffer-exchanged into 5 mM $NaPO_4$ of pH 7.2 using Amicon centrifugal filter units (100-kDa MWCO) immediately before circular dichroism (CD) measurements. A Chirascan CD spectrometer (Applied Photophysics, Leatherhead, UK) was used to record the CD spectra from 180 to 260 nm in a 1-mm quartz cuvette. A bandwidth of 1 nm and a time constant of 1 s were used.

Table 2
Yield and impurity levels achieved using classical and sequential PEG precipitation. mAb precipitate was recovered by centrifugation in both precipitation approaches. Data are given as mean ± standard deviation in the case of triplicate analysis.

Yield (%)	Classical PEG precipitation	Sequential PEG precipitation
mAb1	90 ± 2	92 ± 2
mAb2	88 ± 2	86 ± 3
mAb3	91 ± 1	94 ± 2
DNA (ng/ml)		
mAb1	5113 ± 261	29 ± 4
mAb2	2155 ± 152	18 ± 3
mAb3	678 ± 89	28 ± 7
HCP (ppm)		
mAb1	54,077 ± 4512	15,072 ± 3164
mAb2	37,399 ± 2353	10,042 ± 1571
mAb3	19,953 ± 2691	10,992 ± 3405
HMWI (%)		
mAb1	1.2 ± 0.1	0.9 ± 0.1
mAb2	4.0 ± 0.1	3.9 ± 0.2
mAb3	3.9 ± 0.2	3.8 ± 0.1

3. Results and discussion

3.1. Contaminant removal at low pH

The precipitation screenings were performed in 96-well Deep-Well plates to keep the sample consumption to a minimum while also minimizing errors due to small volumes. PEG concentration and pH have been used as sole screening parameters for both precipitation steps. The results of the screening experiments for both precipitation reactions, the impurity as well as the mAb precipitation, are shown in Fig. 2A and B, respectively.

For the impurity precipitation, a pH range of 3.5–5.5 was screened. The soluble DNA concentration is significantly reduced by pH reduction alone (Fig. 2A). Addition of PEG6000 at low concentrations is sufficient to reduce the DNA concentration to minimal levels. DNA precipitation was most efficient at pH 4.0–4.5. However, at a pH above or below this level, significantly higher PEG6000 concentrations were needed to achieve DNA reduction to similar levels. HCPs were also removed to some extent due to the pH reduction, but addition of PEG6000 up to 7 w/v% did not reduce them further (data not shown). Higher concentrations of PEG resulted in significant loss of mAbs due to precipitation. From the screening results, a pH of 4.0 and 3 w/v% PEG6000 were selected as the optimal conditions for impurity removal for all three antibodies. This step appears to be a promising platform approach, as the DNA of different cell lines is likely to behave very similar, unrelated to the product.

For mAb precipitation in the second step, as expected, the lowest PEG concentration needed for efficient precipitation was found to be close to the respective *pI* for all three antibodies (Fig. 2B). As its feasibility was proven, sequential PEG precipitation was compared with classical PEG precipitation, that is, mAb precipitation without prior impurity precipitation (see Table 2). The second precipitation step did not deteriorate the final mAb yield. DNA is reduced to very low levels in the sequential approach while being undesirably coprecipitated in classical PEG precipitation. HCP removal is also enhanced by a factor of 1.8–3.7 in the sequential approach depending on the mAb. The HMWI levels were similar for both precipitation methods.

The dosage time of the PEG stock solution to reach the final PEG concentration was investigated. The direct addition of solid PEG flocs was also investigated. Here, the PEG flocs dissolved within about 6 min under stirring. Single-step PEG addition and solid PEG addition show similar equilibrium yields, whereas slowly adding PEG over 10 min significantly reduced the mAb yield (see Fig. 3).

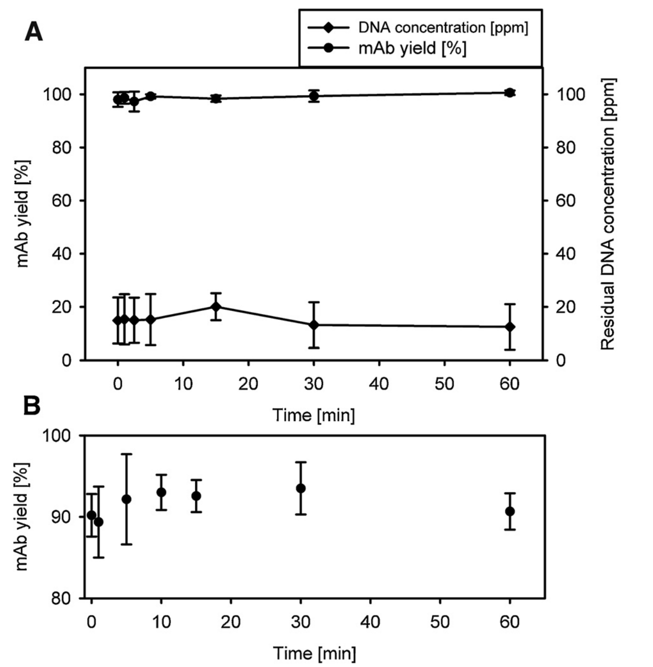

Fig. 4. (A) Speed of DNA precipitation and mAb yield in the low-pH step (pH4/3 w/v% PEG6000) and (B) mAb yield in the second step (pH 7.5/14 w/v% PEG) for mAb3 as a function of incubation time. Point 0 min was defined as the time point at which the sample and PEG stock solution were completely mixed.

These findings indicate that continuous precipitation with only one point of PEG addition, that is, reaching the final PEG concentration immediately, is not only possible but also favorable, as higher yields are obtained by rapid addition of PEG.

The feasibility of precipitation in the continuous mode is also significantly determined by a second factor, the speed of the precipitation reaction. Fast precipitation kinetics directly translate into short required residence times and thus to short tubular reactor lengths. The speed of DNA precipitation in the low-pH step and mAb precipitation in the second step was investigated for mAb3. In both cases, the precipitation reactions were very fast and completed after the sample and PEG stock solution were mixed completely (see Fig. 4A). These data are in agreement with visual examination. Turbidity increased almost instantaneously after adding PEG6000. The mAb3 yield was not compromised even at long residence times at low pH (Fig. 4B).

Based on the results of the kinetics experiments, a residence time of 15 min was selected for the tubular reactors of both precipitation steps. A simple tubular reactor was set up with helical-type static mixers, and both precipitation reactions were performed continuously for mAb3. Impurities were removed by normal-flow filtration between the two precipitation steps.

Samples were drawn during the continuous operation of the IgG precipitation (second step). The mAb precipitate was recovered by centrifugation, and the redissolved samples were analyzed for mAb yield and impurity (DNA and HCP) levels (see Fig. 5). The process was performed at steady state for 2 h, with no drift of mAb yield, HCP, or DNA.

3.2. Continuous PEG precipitation

Continuous precipitation performed almost identically to batch precipitation (Table 3). The mAb yield was above 90% in both operating modes. The concentrations of HCPs and DNA could be reduced to about 10,000 and 20 ppm, respectively. The HMWI levels were not altered by the continuous operating mode.

Table 3
Comparison of batch and continuous PEG precipitation of mAb3 using centrifugation for precipitate recovery. Data are given as mean ± standard deviation in the case of a triplicate analysis.

	Batch precipitation	Continuous precipitation
Yield (%)	94 ± 2	92 ± 1
HCP (ppm)	10,992 ± 3405	10,541 ± 2949
DNA (ppm)	17 ± 3	22 ± 5
HMWI (%)	3.8 ± 0.1	3.8 ± 0.2

Table 4
HF membrane cartridge performance test: Normalized water and 14 w/v% PEG6000 permeability.

Pore size (μm)	Normalized water permeability (LMH)	Normalized PEG permeability (LMH)
0.1	706	120
0.2	3362	883
0.45	6472	1253

3.3. Precipitate recovery by TFF

HF microfiltration membranes were selected, due to their proven suitability in other fields with similar processes such as cell harvest and cell retention in perfusion culture [30,31]. Three different pore sizes were tested in terms of their normalized water and normalized PEG fluxes. Fluxes of 14 w/v% PEG6000 solution in 50 mM sodium phosphate and 150 mM NaCl, at pH 7.5, that is, the PEG concentration used during precipitation, can be used as a model of the actual sample matrix. The increased viscosity increases the pressure and lowers the transmembrane flux in the microfiltration process. Permeate flux versus transmembrane pressure (TMP) data were collected at different feed flow rates, which were linearly fitted and extrapolated to the permeability of the respective solution at a TMP of 1 bar. For both solutions tested, the HF membrane with the smallest pore size (0.1 μm) yielded much lower fluxes than the HF membrane with bigger pore size (Table 4). Interestingly, the water flux of the HF membrane with 0.45-μm pore size is almost double that of the 0.2-μm HF membrane while the flux for the 14 w/v% PEG 6000 solution was only about 40% higher (883 l/(m^2 h) or LMH vs. 1253 LMH). Based on these results, the HF with 0.1-μm pore size was ruled out due to the insufficient fluxes resulting in low throughput and long runtimes or an excessively large filter area.

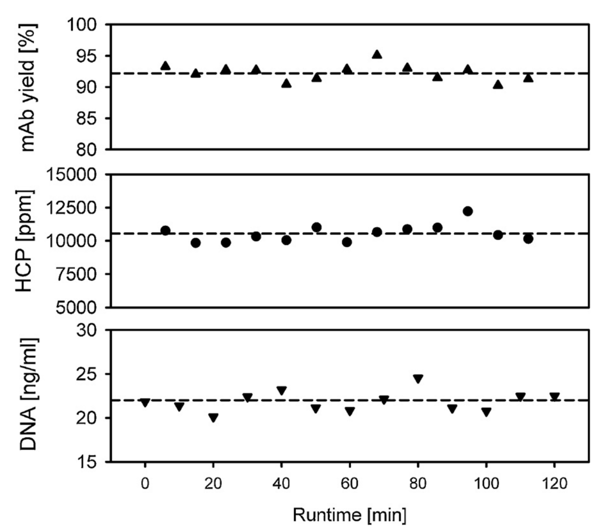

Fig. 5. Critical quality attributes during steady-state operation of the continuous PEG precipitation of mAb3. Precipitate recovery was performed by centrifugation.

TFF was performed in the constant flux mode, as it has been reported to minimize fouling [32,33]. The HF membrane with 0.45-μm pore size was ruled out after initial precipitate concentration trials, as it showed premature blocking of the HF membrane. Differences in HF membrane structure as well as pore structure are the most likely causes for the different behavior. Therefore, concentration and diafiltration runs were performed with HF cartridges with 0.2-μm pore size. The TMP stayed constant up to a concentration factor of about 16 and then started to increase linearly from about 0.09–0.32 bar between a concentration factor of 16–50. The concentration was finished at a concentration factor of 50 (equaling a mAb precipitate concentration of 100 mg/ml) at which point a retentate pressure of 1.25 bar was reached. The TMP decreased slightly during the diafiltration step to about 0.30 bar. The potential loss of the protein precipitate via the passage through the membrane into the permeate was evaluated by drawing permeate samples during precipitate concentration and diafiltration. All samples appeared clear on visual examination with no turbidity at 600 nm, indicating that the entire precipitate is rejected by the membrane.

Precipitate recovery by TFF and centrifugation was compared (see Table 5). Both methods attained the same mAb yield. TFF gave a total yield of 92%, comprising step yields of 94% for the precipitation reaction itself and 98% for the recovery from the TFF system. Residual DNA and HCPs were lower for the TFF recovery; this was most likely because washing the precipitate flocs by diafiltration is more efficient than washing the pelleted precipitate. An increase in HMWI levels during TFF recovery was observed, which can be attributed to the shear exposure in the HF cartridge. The shear rate in a circular duct can be calculated (see Eq. (1)), where Q is the volumetric flow rate, r_{fiber} is the radius of the HFs, and n is the number of fibers in the cartridge [31]:

$$\text{Shear rate} = \frac{4 \times Q}{r_{\text{fiber}}^3 \times n \times \pi} \tag{1}$$

The shear rate in the HF cartridge under the selected operating conditions (feed flow: 675 LMH, permeate flow: 75 LMH) was 1620 s^{-1}, as calculated using Eq. (1).

MAb precipitate was recirculated through the TFF system to investigate the effect of residence time (or cumulative shear exposure) on the HMWI levels after precipitate dissolution. HMWI increased from 3.6% to 5.4% after 6 h of circulation through the system at a feed flow rate of 675 LMH (see Fig. 6).

Circular dichroism (CD) spectroscopy has been extensively used to detect conformational changes in proteins [34–36]. It is used to detect potential changes in the secondary structure due to the interim phase change of the mAb or the recovery method. Samples purified by precipitation and recovered by either centrifugation or by TFF did not reveal any changes in secondary structure when compared to reference material purified by protein A affinity chromatography.

3.4. Dissolution

The dissolution of the concentrated precipitate was tested using four different buffer substances. Buffer systems spanning a wide pH range with conductivities commonly used in ion exchange and multimodal chromatography, both likely to be subsequent polishing steps in a full purification process, were selected (Fig. 7).

Table 5
Comparison of two different precipitate recovery methods (TFF and centrifugation) for mAb3. TFF: The mAb was concentrated to 100 mg/ml prior to dissolution. In both recovery methods, the mAb was dissolved to about 10 mg/ml. Data are given as mean ± standard deviation for triplicate analysis.

	TFF	Centrifugation
Yield (%)	92 ± 3	94 ± 2
DNA (ng/ml)	12 ± 6	17 ± 3
HCP (ppm)	7290 ± 1591	10,992 ± 3405
HMWI (%)	4.5 ± 0.2	3.8 ± 0.1

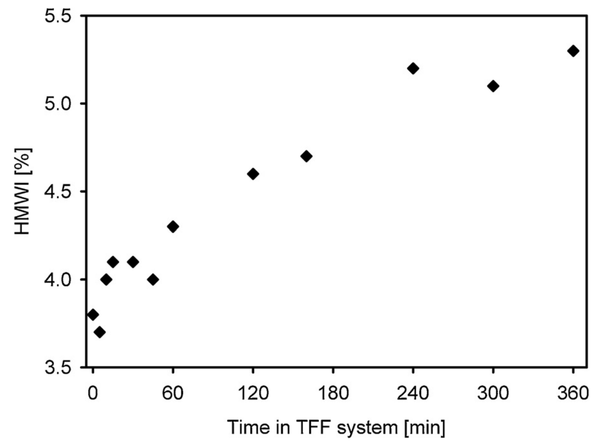

Fig. 6. Influence of residence time in the TFF system at a shear rate of 1619 s^{-1} on HMWI levels after resolubilization. The system was filled with mAb3 precipitate at a concentration of 2 mg/ml and recirculated through the system without concentration.

The mAb precipitate was concentrated to about 90 mg/ml in a 0.2-μm HF cross-flow filtration module and subsequently diafiltered fourfold using 14 w/v% PEG6000 in 25 mM sodium phosphate and 150 mM NaCl at a pH of 7.5 prior to dissolution.

For all tested buffers and all conductivities, the mAb yield was above 90%. The results from the HCP reduction analysis also show no effect on the dissolution buffer. HCPs can be reduced to below 10,000 ppm for all tested buffers. The HMWI content of the dissolved precipitate was similar in all investigated buffers (3.7–4.0%). Only 2.5 mS/cm of Tris–HCl of pH 8.1 shows a markedly lower HMWI content than the other buffers. This HMWI decrease was not observed for Tris–HCl buffers of higher conductivity. Knevelman et al. have made a similar observation [1]. They found that a pH above the *pI* resulted in the highest recovery after resolubilization of a PEG precipitated IgG_4. However, they did not investigate the effect of salt concentration or measured HMWI levels.

In general, it is desirable to dissolve the mAb in as high a concentration as possible to minimize the solution volume. The concentration of residual PEG after resolubilization of the precipitated mAb by TFF is directly correlated with the dilution factor during dissolution. After concentration by TFF, the mAb precipitate is dissolved by diluting the precipitant (PEG). The lowest possible dissolution factor was determined by dissolving highly concentrated mAb3 precipitate to different residual PEG6000 concentrations. A linear relationship between dilution factor or residual PEG concentration and soluble mAb concentration was obtained up to a residual PEG concentration of 3 w/v% (black line in Fig. 8). Below this PEG concentration, the mAb completely dissolves and the equilibrium yield is achieved (red line in Fig. 8). Above this PEG concentration, increasing amounts of the mAb precipitate remain undissolved, resulting in lower soluble mAb concentrations and accompanying decrease in yield.

4. Conclusion and outlook

A sequential PEG precipitation-based capture step for recombinant antibodies was developed in this study. The process was shown to be feasible for three different antibodies with varying titers (2.3–4.9 mg/ml) and *pI* (7.4–9.3) in the batch mode. The parameters and subunit operations essential to optimizing processes for successful implementation were studied. Apart from the actual precipitation, we investigated the recovery of antibody pre-

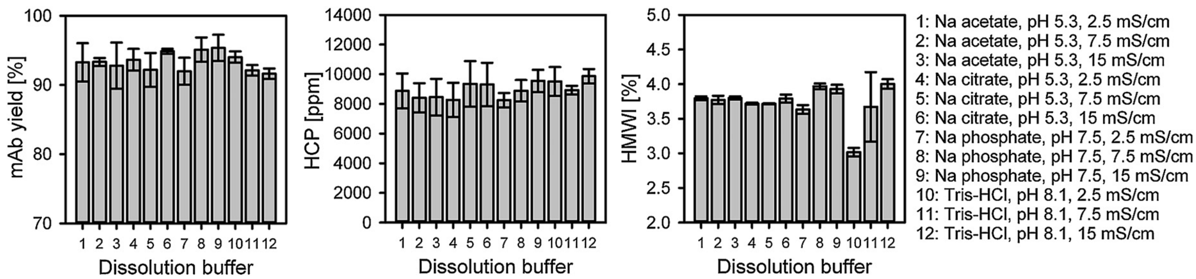

Fig. 7. Twelve different buffers were tested for precipitate dissolution. Approximately 90 mg/ml of the mAb precipitate was diluted in the respective buffers to reach a final dissolved mAb concentration of 9 mg/ml.

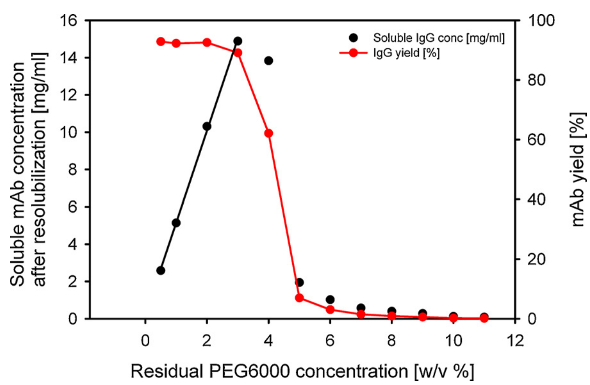

Fig. 8. Soluble mAb concentration and yield after dissolution of mAb precipitate as a function of residual PEG6000 concentration. The mAb precipitate was concentrated to about 100 mg/ml by TFF and diafiltered into 50 mM sodium phosphate buffer, 150 mM NaCl, and 14 w/v% PEG6000, at pH 7.5 prior to the dissolution. PBS was used a resolubilization buffer. For interpretation of the references to colour in the figure, the reader is referred to the web version of this article.

cipitate as well as its resolubilization. Both aspects that are often neglected in precipitation research.

A low-pH impurity removal step, the initial step of sequential precipitation, is a potential platform approach for reducing initial impurity levels. Technology platforms are in great demand, as they are cost-effective and time-saving for developing industrial processes [37]. Transferring the precipitation reaction to the continuous operation mode was demonstrated in a simple, single-use tubular reactor. Steady-state operation was achieved. Performance in terms of yield, DNA, and HCP reduction were very similar for both batch and continuous operation.

An alternative precipitate recovery step was developed. Concentration and diafiltration of the precipitate particles were found to replace the commonly used centrifugation method. In this approach, the individual precipitate flocs are preserved, resulting in rapid dissolution of the precipitate. Increase in HMWI during precipitate recovery by TFF can be minimized by reducing the time to shear exposure. This can be accomplished by either increasing the HF cartridge size and/or using lower concentration factors. Higher titers further help overcome this limitation. As precipitates readily dissolve in various buffer systems, which are used as loading buffers in mixed-mode or ion exchange chromatography, this step can be coupled with subsequent purification steps. Steps not contributing to the purity such as ultra-/diafiltration, required to condition the solution for a next purification step, can be eliminated. The slightly higher viscosity due to residual PEG at low concentrations present after mAb precipitate resolubilization does not affect subsequent unit operations. Residual PEG (3350) has been shown not to interfere with subsequent cation exchange chromatographic steps [2,16]. CD spectroscopy did not reveal any changes in the secondary structure of the mAb compared with the chromatographically purified reference material, neither due to the phase change during precipitation nor due to the recovery by TFF.

The sequential precipitation presented has several advantages over the classical PEG precipitation approach, without further complicating the process. Combined with a TFF-based mAb precipitate recovery, this capture step unites advantages with regard to the use of single-use equipment and continuous processing.

Conflict of interest

The authors declare no conflict of interest.

Acknowledgments

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References

[1] C. Knevelman, J. Davies, L. Allen, N.J. Titchener-Hooker, High-throughput screening techniques for rapid PEG-based precipitation of IgG4 mAb from clarified cell culture supernatant, Biotechnol. Prog. 26 (2010) 697–705.
[2] M. Kuczewski, E. Schirmer, B. Lain, G. Zarbis-Papastoitsis, A single-use purification process for the production of a monoclonal antibody produced in a PER.C6 human cell line, Biotechnol. J. 6 (2011) 56–65.
[3] S. Ghose, T.M. McNerney, B. Hubbard, pH transitions in Ion-exchange systems: role in the development of a cation-exchange process for a recombinant protein, Biotechnol. Prog. 18 (2002) 530–537.
[4] A.S. Rathore, H. Agarwal, A.K. Sharma, M. Pathak, S. Muthukumar, Continuous processing for production of biopharmaceuticals, Prep. Biochem. Biotechnol. 45 (2015) 836–849.
[5] N. Hammerschmidt, B. Hintersteiner, N. Lingg, A. Jungbauer, Continuous precipitation of IgG from CHO cell culture supernatant in a tubular reactor, Biotechnol. J. 10 (2015) 1196–1205.
[6] A. Tscheliessnig, P. Satzer, N. Hammerschmidt, H. Schulz, B. Helk, A. Jungbauer, Ethanol precipitation for purification of recombinant antibodies, J. Biotechnol. 188C (2014) 17–28.
[7] A.M. Azevedo, A.G. Gomes, P.A.J. Rosa, I.F. Ferreira, A.M.M.O. Pisco, M.R. Aires-Barros, Partitioning of human antibodies in polyethylene glycol–sodium citrate aqueous two-phase systems, Sep. Purif. Technol. 65 (2009) 14–21.
[8] P.A. Rosa, I.F. Ferreira, A.M. Azevedo, M.R. Aires-Barros, Aqueous two-phase systems: a viable platform in the manufacturing of biopharmaceuticals, J. Chromatogr. A 1217 (2010) 2296–2305.
[9] R. Sommer, A. Tscheliessnig, P. Satzer, H. Schulz, B. Helk, A. Jungbauer, Capture and intermediate purification of recombinant antibodies with combined precipitation methods, Biochem. Eng. J. 93 (2015) 200–211.
[10] F. Capito, J. Bauer, A. Rapp, C. Schroter, H. Kolmar, B. Stanislawski, Feasibility study of semi-selective protein precipitation with salt-tolerant copolymers for industrial purification of therapeutic antibodies, Biotechnol. Bioeng. 110 (2013) 2915–2927.
[11] V. Boeris, C. Cassane, J. Wagner, G. Pico, The formation of non-soluble complexes between polyethyleneimine-anions and their potential use to isolate enzymes, Colloids Surf. B Biointerfaces 82 (2011) 354–358.

[12] W. Marek, R. Muca, S. Woś, W. Piątkowski, D. Antos, Isolation of monoclonal antibody from a Chinese hamster ovary supernatant. I: Assessment of different separation concepts, J. Chromatogr. A 1305 (2013) 55–63.
[13] S.L. Sim, T. He, A. Tscheliessnig, M. Mueller, R.B. Tan, A. Jungbauer, Branched polyethylene glycol for protein precipitation, Biotechnol. Bioeng. 109 (2012) 736–746.
[14] R. Sommer, P. Satzer, A. Tscheliessnig, H. Schulz, B. Helk, A. Jungbauer, Combined polyethylene glycol and $CaCl_2$ precipitation for the capture and purification of recombinant antibodies, Process Biochem. 49 (2014) 2001–2009.
[15] Y. Zhao, L. Kang, S. Gao, X. Gao, W. Xin, J. Wang, PEG precipitation coupled with chromatography is a new and sufficient method for the purification of botulinum neurotoxin type B, PLoS One 7 (6) (2012) e39670.
[16] G. Giese, A. Myrold, J. Gorrell, J. Persson, Purification of antibodies by precipitating impurities using polyethylene glycol to enable a two chromatography step process, J. Chromatogr. B 938 (2013) 14–21.
[17] P. Satzer, A. Tscheliessnigg, R. Sommer, A. Jungbauer, Separation of recombinant antibodies from DNA using divalent cations, Eng. Life Sci. 14 (2014) 477–484.
[18] Y. Brodsky, C. Zhang, Y. Yigzaw, G. Vedantham, Caprylic acid precipitation method for impurity reduction: an alternative to conventional chromatography for monoclonal antibody purification, Biotechnol. Bioeng. 109 (2012) 2589–2598.
[19] S. Tomic, L. Besnard, B. Fürst, R. Reithmeier, R. Wichmann, P. Schelling, C. Hakemeyer, Complete clarification solution for processing high density cell culture harvests, Sep. Purif. Technol. 141 (2015) 269–275.
[20] Y. Kang, J. Hamzik, M. Felo, B. Qi, J. Lee, S. Ng, G. Liebisch, B. Shanehsaz, N. Singh, K. Persaud, D.L. Ludwig, P. Balderes, Development of a novel and efficient cell culture flocculation process using a stimulus responsive polymer to streamline antibody purification processes, Biotechnol. Bioeng. 110 (2013) 2928–2937.
[21] T. McNerney, A. Thomas, A. Senczuk, K. Petty, X. Zhao, R. Piper, J. Carvalho, M. Hammond, S. Sawant, J. Bussiere, PDADMAC flocculation of Chinese hamster ovary cells: enabling a centrifuge-less harvest process for monoclonal antibodies, mAbs 7 (2015) 413–427.
[22] K.A. Kantardjieff, B. Rupp, Protein isoelectric point as a predictor for increased crystallization screening efficiency, Bioinformatics 20 (2004) 2162–2168.
[23] K.M. Champion, D. Arnott, W.J. Henzel, S. Hermes, S. Weikert, J. Stults, M. Vanderlaan, L. Krummen, A two-dimensional protein map of Chinese hamster ovary cells, Electrophoresis 20 (1999) 994–1000.
[24] T. Vicente, S. Burri, S. Wellnitz, K. Walsh, S. Rothe, J. Liderfelt, Fully aseptic single-use cross flow filtration system for clarification and concentration of cytomegalovirus-like particles, Eng. Life Sci. 14 (2014) 318–326.
[25] B. Cunha, C. Peixoto, M.M. Silva, M.J.T. Carrondo, M. Serra, P.M. Alves, Filtration methodologies for the clarification and concentration of human mesenchymal stem cells, J. Membr. Sci. 478 (2015) 117–129.
[26] A. Polson, G.M. Potgieter, J.F. Largier, G.E. Mears, F.J. Joubert, The fractionation of protein mixtures by linear polymers of high molecular weight, Biochim. Biophys. Acta 82 (1964) 463–475.
[27] I.R. Juckes, Fractionation of proteins and viruses with polyethylene glycol, Biochim. Biophys. Acta 229 (1971) 535–546.
[28] K.C. Ingham, Precipitation of proteins with polyethylene glycol, in: P.D. Murray (Ed.), Methods Enzymol, vol. 182, Academic Press, 1990, pp. 301–306.
[29] A. Tscheliessnig, A. Jungbauer, High-performance monolith affinity chromatography for fast quantitation of immunoglobulin G, J. Chromatogr. A 1216 (2009) 2676–2682.
[30] V. Warikoo, R. Godawat, K. Brower, S. Jain, D. Cummings, E. Simons, T. Johnson, J. Walther, M. Yu, B. Wright, J. McLarty, K.P. Karey, C. Hwang, W. Zhou, F. Riske, K. Konstantinov, Integrated continuous production of recombinant therapeutic proteins, Biotechnol. Bioeng. 109 (2012) 3018–3029.
[31] M.-F. Clincke, C. Mölleryd, Y. Zhang, E. Lindskog, K. Walsh, V. Chotteau, Very high density of CHO cells in perfusion by ATF or TFF in WAVE bioreactor™. Part I. Effect of the cell density on the process, Biotechnol. Prog. 29 (2013) 754–767.
[32] B. Maiorella, G. Dorin, A. Carion, D. Harano, Crossflow microfiltration of animal cells, Biotechnol. Bioeng. 37 (1991) 121–126.
[33] T. Kluge, C. Rezende, D. Wood, G. Belfort, Protein transmission during Dean vortex microfiltration of yeast suspensions, Biotechnol. Bioeng. 65 (1999) 649–658.
[34] S.M. Kelly, T.J. Jess, N.C. Price, How to study proteins by circular dichroism, Biochim. Biophys. Acta 1751 (2005) 119–139.
[35] S. Fiedler, L. Cole, S. Keller, Automated circular dichroism spectroscopy for medium-throughput analysis of protein conformation, Anal. Chem. 85 (2012) 1868–1872.
[36] A. Seidl, O. Hainzl, M. Richter, R. Fischer, S. Böhm, B. Deutel, M. Hartinger, J. Windisch, N. Casadevall, G. London, I. Macdougall, Tungsten-induced denaturation and aggregation of epoetin alfa during primary packaging as a cause of immunogenicity, Pharm. Res. 29 (2012) 1454–1467.
[37] S.S. Farid, Process Economic Drivers in Industrial Monoclonal Antibody Manufacture. Process Scale Purification of Antibodies, John Wiley & Sons, Inc., 2008, pp. 239–261.